Number theory

onsdag den 23. oktober 2024 13.17

Antag, at $a|b$ og $a|c$. Vis, at $a|(b+c)$

Tag $k, l \in \mathbb{Z}$ så
$ak = b$ og $al = c$

nu er
$b+c = ak + al = a(k+l)$, så $a|(b+c)$

Antag, at $a|b$ & $a|c$. Vis, at $a|c$
$a|(bk+cl)$

Hvis $a \equiv b$ & $c \equiv d$

$a+c = b+d$
$ac = bd$

$(a+b) \bmod m = (a \bmod m) + (b \bmod m)$

$(a \cdot b) \bmod m = (a \bmod m) \cdot (b \bmod m)$

Bevis

$a \equiv (a \bmod m) \bmod m$
$b \equiv (b \bmod m) \bmod m$
$a+b = (a \bmod m) + (b \bmod m) \bmod m$

Aritmetikkens fundamentalsætning

$n = p_1^{a} p_2^{a} \cdots p_k^{a}$

der er uendeligt mange primtal

Bevis:

antag at der er endeligt mange primtal: $p_1, p_2, \ldots p_n$. Betragt

$Q = p_1 p_2 \cdots p_n + 1$

Fra antagelsen er Q ikke et primtal (fundamentalsætningen)

Fx vil $p_j | Q$ for et eller andet $j = 1, 2, \ldots n$.

men vi har også at $p_j | p_1 p_2 \cdots p_n$,

så $p_j | \underbrace{Q - p_1 p_2 \cdots p_n}_{=1}$

Det er ikke muligt □

Quotient & Remainder

$a = dq + r$
$a \text{ div } d = q$
$a \bmod d = r$

Kongruens

$1 \equiv 13 \bmod 12$
$a \equiv b \pmod m$
$a \bmod m = b \bmod m$
there exists $k \in \mathbb{Z}$ with $a = b + km$
DVS $m | (a-b)$

Ækvivalensrelation!

$\gcd(a,b)\{d \mid d|a \wedge d|b\}$

$\gcd(18, 24) = 2 \cdot 3 = 6$

$18 = 2 \cdot 3 \cdot 3$
$24 = 2 \cdot 2 \cdot 2 \cdot 3$

$\text{lcm}(a,b) = \{m \mid a|m \wedge b|m\}$

$\text{lcm}(18, 24) = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 = 72$

$18 = 2 \cdot 3 \cdot 3$
$24 = 2 \cdot 2 \cdot 2 \cdot 3$

$\gcd(1000, 350)$

$2^3 \cdot 5^3 \cdot 7^0$
$2^1 \cdot 5^2 \cdot 7^1$

$2^1 \cdot 5^2 \cdot 7^0 = 50$

$\text{lcm}(63, 35)$

$3^2 \cdot 7^1 \quad 3^2 \cdot 5^1 \cdot 7^1 = 350$
$3 \cdot 5^1 \cdot 7^1$

Euklids algoritme

$\gcd(\underset{a}{287}, \underset{b}{91}) = \gcd(91, 14) = \gcd(14, 7) = \gcd(7, 0) = 7$

$287 = \underline{91} \cdot 3 + \underline{14}$

$91 = \underline{14} \cdot 6 + \underline{17}$

$14 = 7 \cdot 2 + \underline{\underline{0}}$

```
def gcd(a:int, b:int) -> int:
    """ returns the greatest common divisor
        requires a>b.
    """
    x, y = a, b
    while y != 0
        (x, y = y, x%y)
    return x
```

Let $a, b \in \mathbb{Z}, c \in \mathbb{Z} \setminus \{0\} \wedge m \in \mathbb{Z}^+$.
if $ac \equiv bc \pmod m \wedge \gcd(c, m) = 1$
then $a \equiv b \pmod{}$

Let $a, b, c \in \mathbb{Z}^+$ with $\gcd(a,b) = 1 \wedge a|bc$
then $a|c$

Congruences

$ax \equiv b \pmod m$
$4x \equiv 5 \bmod 10$
$x = 3 + 5k$

Bézouts Sætning

Let $a, b \in \mathbb{Z}^+$
there exist $s, t \in \mathbb{Z}$ with $\gcd(a,b) = sa + tb$

if $m | (a-b)c$, $\gcd(c, m) = 1$
then $m | a-b$, $\Rightarrow$ $a \equiv b \bmod m$

if a has a $\bar{a}$
then $\bar{a}ax \equiv \bar{a}b$, $x \equiv \bar{a}b \pmod m$

Multiplicative Inverse

Let $a, m \in \mathbb{Z}$ with $m \geq 2$.
if $\gcd(a, m) = 1$
there exits precisely one $\bar{a} \in \mathbb{Z}_m$ with
$\bar{a} \cdot a \equiv 1 \bmod m$

Bevis

Because $\gcd(a, m) = 1$
And Bezout says that $s, t \in \mathbb{Z}$
so $1 = sa + tm$
therefore $mt = 1 - sa$.
therefore $m | 1 - sa$
therefore $sa \equiv 1 \bmod m$
now $\bar{a} = s$ is a multiplicative iverse for $a \bmod m$

if $\gcd(a,b) = 1$, there exists $\bar{a}$ and a uniqe solution to $\mathbb{Z}_m$
if $\gcd(a,b) \neq 1$, $\bar{a}$ does not exist and there may be a solution to $\mathbb{Z}_m$

$\underbrace{\bar{a} \cdot a}_{1} x \equiv b\bar{a} \bmod m$

Chinese Remainder Theorem

$\gcd(m_k, m_{k+1}) = 1$

Betragt
$x \equiv 2 \bmod 3$
$x \equiv 3 \bmod 5$
$x \equiv 2 \bmod 7$

find coefficients

$b_1 \begin{cases} 1 \bmod 3 \\ 0 \bmod 5 \\ 0 \bmod 7 \end{cases} \quad b_2 \begin{cases} 0 \bmod 3 \\ 1 \bmod 5 \\ 0 \bmod 7 \end{cases} \quad b_3 \begin{cases} 0 \bmod 3 \\ 0 \bmod 5 \\ 1 \bmod 7 \end{cases}$

$x = 2b_1 + 3b_2 + 2b_3$

$b_1 = 70 = 23 \cdot 3 + 1 = 14 \cdot 5 + 0 = 10 \cdot 7 + 0$

$b_2 = 21 = 7 \cdot 3 + 0 = 4 \cdot 5 + 1 = 3 \cdot 7 + 0$

$b_3 = 15 = 5 \cdot 3 + 0 = 3 \cdot 5 + 0 = 2 \cdot 7 + 1$